NNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME

CLASS

INDEX NUMBER

CHEMISTRY

9647/02

Paper 2 Structured Questions

2 Sept 2015

2 hours

Candidates answer on the Question Paper

Additional Materials: *Data Booklet*

READ THESE INSTRUCTIONS FIRST

Write your index number, name and civics group.
Write in dark blue or black pen.
You may use pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the space provided.
A Data Booklet is provided.

You are advised to show all working in calculations.
You are reminded of the need for good English and clear presentation in your answers.
You are reminded of the need for good handwriting.
Your final answers should be in 3 significant figures.

You may use a calculator.

The number of marks is given in brackets [ ] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	**12**
2	**17**
3	**12**
4	**16**
5	**15**
Significant figures	
Handwriting	
Total	**72**

This document consists of **16** printed pages and **2** blank pages.

For Examiner's Use

Answer ALL questions on the spaces provided.

1 An experiment can be set up to identify an unknown metal M and the charge of its ions, M^{n+}. This is achieved by investigating how the cell potential containing this metal M in contact with an aqueous solution of its ions, $M^{n+}(aq)$ (where n = 1, 2 or 3), changes as $M^{n+}(aq)$ is diluted.

It is known that the cell potential of a cell, E_{cell}, is related to the standard electrode potential, $E_{cell}^{\ominus}$, by the following equation.

$$E_{cell} = E_{cell}^{\ominus} - \frac{0.06 \log[M^{n+}(aq)]}{n}$$

Since a standard hydrogen half-cell was not available, a standard half-cell consisting of silver in contact with a 1.0 mol dm^{-3} solution of silver nitrate was used to connect to the half-cell with metal M, in contact with $M^{n+}(aq)$.

(a) Draw a labelled diagram you would set up in order to measure the cell potential of this electrochemical cell.

[2]

(b) With relevant data from the *Data Booklet*, suggest whether the M^{n+}/M half-cell is the oxidation or reduction half-cell in **(a)**.

Explain your answer.

Type of half-cell: ……………………………………………………………………………...

Explanation: ……………………………………………………………………………………..

……………………………………………………………………………………………………..

[1]

For Examiner's Use

(c) Write a plan for conducting a series of experiments to investigate how the cell potential changes as the aqueous solution of its ions, $M^{n+}(aq)$, is diluted.

You may assume that you are provided with the following.

- 1.0 mol dm^{-3} solution of silver nitrate, $AgNO_3(aq)$
- 0.5 mol dm^{-3} solution of metal M ions, $M^{n+}(aq)$
- deionised water
- the apparatus normally found in a school or college laboratory

Your plan should include the following.

- details for the preparation of a suitable range of diluted solutions of $M^{n+}(aq)$ of accurate concentrations;
- description of how the electrochemical cell set-up in **(a)** is assembled;
- an outline of how results would be obtained;
- on the grid provided, a sketch of the graph that shows the relationship between log $[M^{n+}(aq)]$ and the cell potential measured, E_{cell};
- an explanation of the shape of this graph;
- how the data obtained from this graph would be used to determine the
  - charge of metal ion, M^{n+}
  - the standard electrode potential for metal, M, $E^{\ominus}(M^{n+}/M)$.

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For Examiner's Use

[9]

[Total: 12]

For Examiner's Use

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For Examiner's Use

2 Nitrogen forms two stable anions, NO_3^- and NO_2^-. Bromine forms four stable anions, BrO^-, BrO_2^-, BrO_3^- and BrO_4^-.

(a) All group II nitrates decompose readily on heating, leaving a white residue.

(i) Write an equation showing the reaction that occurs when $Ca(NO_3)_2$ is heated.

.... [1]

(ii) Explain the variation down the group of the ease of thermal decomposition of Group II metal nitrates.

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.... [2]

(b) Bromate(V), BrO_3^- reacts with acidified hydrogen peroxide, H_2O_2 to form bromine, Br_2.

(i) Write the half-equation for the reduction of bromate(V), BrO_3^- ions.

.... [1]

(ii) Hence, write the overall equation for the reaction between bromate(V) ions and acidified hydrogen peroxide.

.... [1]

(iii) The reaction can be catalysed by transition metal ions.

What feature of transition metals such as iron makes them suitable as homogeneous catalysts in chemical reactions?

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.... [1]

For Examiner's Use

(c) Bromine reacts with hydrogen in the following reaction.

$$H_2 + Br_2 \longrightarrow 2HBr$$

This reaction is thought to occur by the following sequence of reactions.

$$Br_2 \rightleftharpoons 2Br \quad \text{(fast)}$$

$$H_2 + Br \longrightarrow HBr + H \quad \text{(slow)}$$

$$H + Br_2 \longrightarrow HBr + Br \quad \text{(fast)}$$

(i) Explain what is meant by the term *order of reaction*.

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.... [1]

(ii) From the sequence of reactions given, show that the rate equation is:

$$\text{rate} = k\,[H_2]\,[Br_2]^{1/2}$$

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.... [2]

(d) The kinetics of the reaction between bromate(V) and acidified bromide ions can be studied by the "clock" method, using phenol. The equation for the reaction is as follows.

$$BrO_3^-(aq) + 5Br^-(aq) + 6H^+(aq) \longrightarrow 3Br_2(aq) + 3H_2O(l) \quad \text{reaction } \mathbf{I}$$

When small but constant amounts of phenol and methyl orange is added to the reaction mixture, the bromine produced by the reaction will immediately react with the phenol present until all the phenol has been used up.

(i) Write a balanced equation for the reaction of phenol with aqueous bromine.

.... [1]

As soon as the reaction **I** has produced sufficient bromine to react with the fixed amount of phenol, free bromine will appear in the solution. This bromine will immediately bleach the methyl orange solution causing the solution to turn from pink to colourless.

$$Br_2 + \underset{\text{(pink)}}{\text{methyl orange}} \longrightarrow \underset{\text{(colourless)}}{\text{bleached methyl orange}} + 2Br^-$$

(ii) What is the simple relationship between the time taken for the solution to turn from pink to colourless and the initial rate of reaction **I**?

.... [1]

A series of experiments was carried out using different volumes of the reagents, each solution being made up to the same total volume with water.

The following results were obtained.

no.	vol. of 0.50 $mol dm^{-3}$ $NaBrO_3$ / cm^3	vol. of 0.10 $mol dm^{-3}$ NaBr / cm^3	vol. of 1.00 $mol dm^{-3}$ H_2SO_4 / cm^3	vol. of phenol/ cm^3	vol. of methyl orange / cm^3	vol. of water / cm^3	time taken for solution to turn from pink to colourless / s
1	15	10	5	3	2	75	140
2	30	10	5	3	2	60	70
3	15	15	5	3	2	70	94
4	15	10	15	3	2	65	16

(iii) Calculate the initial rate for each of these four experiments and use the results to deduce the order of the reaction with respect to each of the reactants. Explain your reasoning.

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.... [4]

For Examiner's Use

(iv) Hence write the overall rate equation for the reaction, stating the units for the rate constant.

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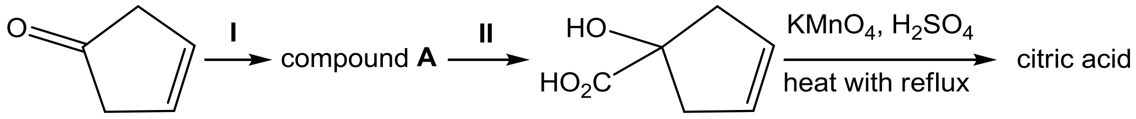
[Total: 17]

For Examiner's Use

3 Citric acid is a commodity chemical and more than a million ton is produced every year by fermentation. It is a natural preservative which is present in citrus fruits.

Citric acid can be synthesised from cyclopent-3-enone by the following route:



(a) **(i)** Suggest the reagents and conditions for steps **I** and **II**.

Step **I**: ……………………………………………………………………………………...

Step **II**: ……………………………………………………………………………………..

[2]

(ii) Draw the structural formula of compound **A** and citric acid.

compound **A**	citric acid

[2]

A carboxylic acid present in unripe fruits is a colourless crystalline solid, **V**, which has the following composition by mass: C, 35.8%; H, 4.5%; O, 59.7%.

(b) **(i)** Given that the M_r of **V** is 134, use this value and the above information to determine the molecular formula of **V**.

Molecular formula of **V** …………………

[2]

A sample of **V** of mass 1.97 g was dissolved in water and the resulting solution titrated with 1.00 mol dm^{-3} NaOH. 29.4 cm^3 were required for complete neutralization.

(c) **(i)** Use these data to deduce the number of carboxylic acid groups present in one molecule of **V**.

[1]

(ii) Compound **V** does not decolourise acidified $KMnO_4$ when heated. Compound **W**, an isomer of compound **V**, can be dehydrated to give a mixture of two isomeric alkenedioic acids, compound **X** and **Y** with molecular formula $C_4H_4O_4$.

Gently heating the anhydrous crystals of compound **Y** produces a neutral compound **Z**, $C_4H_2O_3$, which neither reacts with sodium metal nor gives a precipitate with 2,4-dinitrophenylhydrazine.

Suggest the identities of **V**, **W**, **X**, **Y** and **Z**.

compound **V**	compound **W**
compound **X**	compound **Y**
compound **Z**	

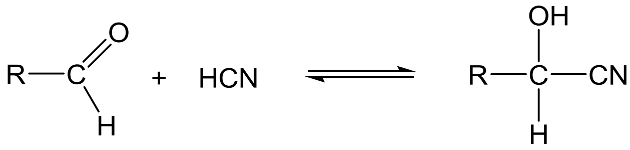
[5]

[Total: 12]

For Examiner's Use

4 **(a)** This question is about a series of experiments conducted on carbon compounds and its derivatives in the laboratory.

The uncatalysed addition reaction of HCN to a carbonyl group to give a cyanohydrin can be represented as a reaction in equilibrium as follows:

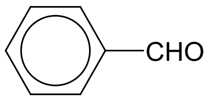
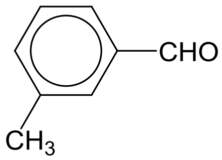
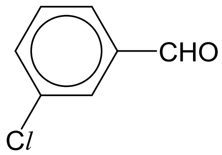
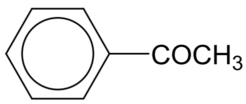


carbonyl compound   cyanohydrin

The K_c expression is given as

$$K_c = \frac{[\text{cyanohydrin}]}{[\text{carbonyl compound}][\text{HCN}]}$$

The value of K_c is determined under identical conditions for the following carbonyl compounds.

no.	carbonyl compound	K_c / mol^{-1} dm^3
1		210
2		175
3		500
4	CH_3COCH_3	30
5		0.8

For Examiner's Use

(i) Explain the differing K_c values for carbonyl compounds 1, 2 and 3.

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.... [3]

(ii) Explain why the value of K_c for carbonyl compound 5 is less than that for 4.

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.... [2]

(b) In another reaction, the thiocyanate ion, SCN^- is used to test for the presence of iron(III) ions.

$$Fe(H_2O)_6^{3+}(aq) + SCN^-(aq) \rightleftharpoons [Fe(SCN)(H_2O)_5]^{2+}(aq) + H_2O(l)$$

(i) Write the K_c expression for the reaction, stating its units.

[2]

(ii) When equal volumes of 8.00×10^{-3} mol dm^{-3} of $FeCl_3$ and 8.00×10^{-3} mol dm^{-3} of KSCN are mixed and allowed to reach equilibrium, the resulting concentration of $[Fe(SCN)(H_2O)_5]^{2+}$(aq) is 3.31×10^{-3} mol dm^{-3}.

Use this data to calculate the value of the equilibrium constant, K_c.

[2]

For Examiner's Use

(iii) The numerical value of K_c for this reaction was found to be 138 when the equilibrium mixture was placed in a hot water bath.

With reference to your answer in **(ii)**, deduce whether the reaction is exothermic or endothermic.

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(c) A student wished to determine the standard enthalpy change of reaction between methane and chlorine to form carbon tetrachloride, $\Delta H_1^\ominus$.

$$CH_4(g) + 4Cl_2(g) \longrightarrow CCl_4(l) + 4HCl(g) \qquad \Delta H_1^\ominus$$

(i) Suggest the conditions required for this reaction to occur.

…………………………………………………………………………………………….. [1]

(ii) A teacher commented that despite having the appropriate conditions for the reaction to take place, it is still not possible to determine $\Delta H_1^\ominus$ directly in the laboratory.

Suggest why this is so.

…………………………………………………………………………………………….. [1]

(iii) The teacher proceeded to provide the following information for the student to determine the enthalpy change of reaction, $\Delta H_1^\ominus$.

$$C(s) + 2H_2(g) \longrightarrow CH_4(g) \qquad \Delta H_2^\ominus$$

$$\frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g) \longrightarrow HCl(g) \qquad \Delta H_3^\ominus$$

By means of an energy cycle or otherwise, determine what other information is needed to calculate a value for $\Delta H_1^\ominus$.

[3]

[Total: 16]

5 **(a)** **(i)** Describe the conditions needed to hydrolyse amides in the laboratory.

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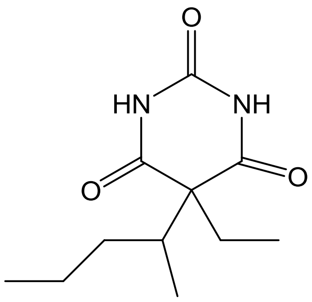
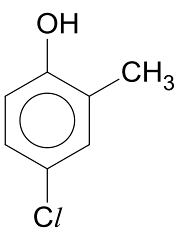
(ii) Predict the products of the hydrolysis of the amide groups in the following drug.



pentobarbital

[2]

(b) Compound **S** can be formed from 2-methylphenol.



S

(i) State the reagents and conditions required for the conversion.

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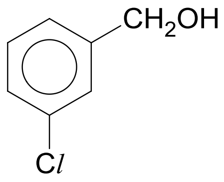
(ii) Describe a simple chemical test that can be used to distinguish **S** from its isomer, **T**. State the reagents and conditions used and the observation for both compounds.



T

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(c) Compound **C** has the molecular formula $C_8H_6O_4$. It is an aromatic compound which contains three functional groups.

Data about the reactions of **C** are given in the table.

reaction	reagent	result
1	neutral $FeCl_3$(aq)	violet complex formed
2	Br_2(aq) in an excess	Br_2 is decolourised; white solid formed with M_r = 323.8
3	2,4-dinitrophenylhydrazine	orange precipitate formed
4	PCl_5	white fumes and a neutral cyclic compound **D** formed with M_r = 148
5	Na	colourless gas evolved.
6	$[Ag(NH_3)_2]^+$(aq)	no silver mirror seen

(i) Name the functional group that reaction **1** shows to be present in **C**.

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(ii) Name the functional group that reaction **3** and **6** show to be present in **C**.

……………………………………………………………………………………………[1]

(iii) 0.1 mol of **C** is reacted with excess sodium metal to produce hydrogen gas. The gas formed occupied a volume of 2.45 dm^3 at room temperature and pressure. Find the number of moles of hydrogen gas formed.

[2]

(iv) Based on reaction **5** and your answer in **(i), (ii)** and **(iii)**, deduce, with reasoning, the identity of the third functional group present in **C**.

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(v) Deduce the molecular formula of the product formed in reaction **2**.

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(vi) Draw the fully displayed structures of **C** and **D**.

[2]

[Total: 15]

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